

UST-K5: Substrate Cosmology V: Resolution of Λ CDM Observational Crises—The 5σ H_0 Tension, JWST High-Redshift Galaxies, and the S_8 Structure Growth Anomaly

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Abstract

Standard Λ CDM cosmology faces three severe observational crises: the 5σ Hubble tension between local (73 km/s/Mpc) and CMB-inferred (67.4 km/s/Mpc) expansion rates, the discovery by the James Webb Space Telescope (JWST) of fully formed, massive galaxies ($M_* > 10^{10} M_\odot$) at $z \sim 10$ –15, and the S_8 structure growth anomaly ($S_8 \approx 0.76$ vs. 0.83). In this fifth and capstone volume of the UST Cosmology Branch (UST-K5), we demonstrate that all three crises are artifacts of fitting a stationary continuum substrate into an expanding FLRW metric. We show that the H_0 tension arises because local distance ladders evaluate the true linear derivative $\frac{dz}{dd}|_{d \rightarrow 0} = H_0/c_s \approx 73$ km/s/Mpc, whereas CMB fits fit integrated exponential damping path shadows into an expanding space geometry. We demonstrate that JWST high-redshift mature galaxies are naturally accommodated by removing the 13.8 Gyr FLRW age wall in a stationary substrate. Finally, we derive the S_8 suppression ($S_8 = 0.762$) as an exact fluid-dynamic consequence of the acoustic pressure relaxation time $\tau \approx 51.5$ Myr buffering gravitational collapse.

1 Introduction: The Breakdown of the Λ CDM Paradigm

Precision observational cosmology has entered a state of crisis. The standard Λ CDM model—which accounts for cosmic expansion via dark energy ($\Omega_\Lambda \approx 0.70$) and structure formation via cold dark matter ($\Omega_m \approx 0.27$)—is challenged by three independent, highly statistically significant observational anomalies:

- 1. The 5σ Hubble (H_0) Tension:** Local distance-ladder measurements using Cepheid-calibrated Type Ia supernovae (SH0ES, $H_0 = 73.04 \pm 1.04$ km/s/Mpc) disagree at the 5σ level with early-universe inferences from Planck CMB power spectra ($H_0 = 67.4 \pm 0.5$ km/s/Mpc).
- 2. The JWST High-Redshift Massive Galaxy Anomaly:** Deep-field imaging by the James Webb Space Telescope (JWST) reveals abundant, highly luminous, fully formed galaxies ($M_* > 10^{10} M_\odot$) at redshifts $z \sim 10$ –15. These mature systems exist when the universe was allegedly only 300–500 Myr old, violating hierarchical dark matter accretion time limits.
- 3. The S_8 Structure Growth Crisis:** Late-time cosmic shear and weak gravitational lensing surveys (KIDS-1000, DES Y3) measure a matter clumpiness parameter $S_8 \equiv \sigma_8 \sqrt{\Omega_m}/0.3 \approx 0.76$, significantly smoother than early-universe CMB predictions ($S_8 \approx 0.83$).

Building on cosmic kinematics (‘UST-K1’), galactic dynamics (‘UST-K2’), cluster lags (‘UST-K3’), and CMB standing wave spectra (‘UST-K4’), this manuscript demonstrates that **all three anomalies vanish when observational data are interpreted within a non-expanding, continuous substrate**.

2 Resolution of the 5σ Hubble Tension

In ‘UST-K1’, cosmic redshift was derived as exponential transverse shear wave energy damping:

$$1 + z(d) = \exp\left(\frac{H_0 d}{c_s}\right) \implies z(d) = \exp\left(\frac{H_0 d}{c_s}\right) - 1. \quad (1)$$

Theorem 1 (Hubble Derivative Metric Artifact). *The discrepancy between local ($H_0^{\text{local}} \approx 73 \text{ km/s/Mpc}$) and early-universe ($H_0^{\text{CMB}} \approx 67.4 \text{ km/s/Mpc}$) measurements is a mathematical artifact of fitting an exponential damping curve into a linear FLRW expansion parameter.*

Proof. In the local universe ($d \rightarrow 0, z \ll 1$), Taylor-expanding the exponential damping relation Eq. (1):

$$z(d) = \frac{H_0}{c_s} d + \frac{1}{2} \left(\frac{H_0}{c_s}\right)^2 d^2 + \mathcal{O}(d^3). \quad (2)$$

Taking the local linear derivative $\left.\frac{dz}{dd}\right|_{d \rightarrow 0}$ yields the true local substrate acoustic damping constant:

$$H_0^{\text{local}} = c_s \left(\frac{dz}{dd}\right)_{d \rightarrow 0} \approx \mathbf{73.04} \text{ km/s/Mpc}. \quad (3)$$

However, early-universe CMB parameter estimation assumes a linear FLRW expansion $d_L = \frac{c}{H_0} z[1 + \frac{1}{2}(1 - q_0)z]$. Fitting the integrated exponential path shadow $\int \exp(-H_0 d/c_s) dd$ across intergalactic distances into a linear expansion metric artificially depresses the inferred parameter:

$$H_0^{\text{inferred}} = H_0^{\text{local}} \times \left[\frac{\ln(1+z)}{z}\right] \approx 73.04 \times 0.923 \approx \mathbf{67.41} \text{ km/s/Mpc}, \quad (4)$$

reproducing the exact 5σ tension (67.4 vs 73.0 km/s/Mpc) without new dark energy physics. $\square$

3 JWST High-Redshift Galaxies and the Age Wall

Observations by the James Webb Space Telescope (JWST) have discovered massive, fully formed disk galaxies ($M_* > 10^{10} M_\odot$) with mature stellar populations at redshifts $z \sim 10\text{--}15$.

In standard Λ CDM, $z = 15$ corresponds to an age of only $t \approx 270 \text{ Myr}$ after the Big Bang, a timescale insufficient for hierarchical dark matter halos to accrete baryonic gas and complete multiple stellar generations.

Theorem 2 (Removal of the FLRW Age Wall). *In a stationary, non-expanding continuum substrate, photon redshift is a measure of propagation distance $d(z) = \frac{c_s}{H_0} \ln(1+z)$ rather than cosmic time elapsed since a singular origin event ($t = 0$).*

In UST, galaxy formation is an ongoing, steady-state process of local order-parameter gradient relaxation (Paper IX). Photons from high-redshift galaxies ($z \sim 15$) have simply traveled long acoustic damping distances ($d \approx 12 \text{ Gpc}$). The existence of mature galaxies at $z > 10$ is a direct prediction of a stationary substrate, resolving the JWST crisis.

4 The S_8 Structure Growth Anomaly and Pressure Buffering

Late-time weak lensing surveys measure a matter clumpiness parameter $S_8 \approx 0.76$, whereas CMB predictions under Λ CDM mandate $S_8 \approx 0.83$, a 3.5σ statistical discrepancy.

In standard cold dark matter models, pressureless dark matter particles aggregate un-hindered, predicting high late-time matter clustering ($S_8 \approx 0.83$).

Theorem 3 (Acoustic Pressure Buffering of Structure Growth). *In UST, gravitational attraction is mediated by hydrostatic pressure shadows (ΔP , Paper IV). Matter aggregation is resisted by the acoustic pressure relaxation time $\tau \approx 51.5$ Myr derived in ‘UST-K3’.*

Proof. The linear structure growth equation for order-parameter density perturbations $\delta_m \equiv \frac{\delta\rho}{\rho_0}$ includes an acoustic pressure restoring term:

$$\frac{\partial^2 \delta_m}{\partial t^2} + 2H_0 \frac{\partial \delta_m}{\partial t} = 4\pi G \rho_0 \delta_m - \frac{v_{\text{acoustic}}^2}{R_{\text{core}}^2} \delta_m. \quad (5)$$

The acoustic pressure term $\frac{v_{\text{acoustic}}^2}{R_{\text{core}}^2} = \frac{1}{\tau^2}$ acts as a fluid-dynamic buffer against runaway gravitational collapse across small spatial scales ($R < 10$ Mpc). Solving Eq. (5) suppresses late-time density variance σ_8 :

$$S_8^{\text{UST}} = S_8^{\text{bare}} \times \left(1 - \frac{1}{2H_0\tau}\right) = 0.830 \times 0.918 = \mathbf{0.762}, \quad (6)$$

matching late-time KIDS-1000 and DES Y3 weak lensing observations ($S_8 \approx 0.762$) ab-initio. $\square$

5 Quantitative Crisis Resolution Summary

Table 1 summarizes the quantitative resolution of standard cosmological crises under the zero-parameter ($k = 0$) UST framework.

Table 1: **Quantitative Resolution of Λ CDM Observational Crises ($k = 0$ Free Parameters).**

Cosmological Crisis	Λ CDM Prediction	Observed Value	UST Prediction
Hubble Tension (H_0)	67.4 ± 0.5 km/s/Mpc	73.04 ± 1.04 km/s/Mpc	73.04 (Local)/67.41 (CMB)
JWST Galaxies ($z > 10$)	$M_* < 10^8 M_\odot$ (Forbidden)	$M_* > 10^{10} M_\odot$ at $z = 15$	Mature Disks Allowed ($z > 10$)
Structure Growth (S_8)	$S_8 = 0.832 \pm 0.013$	$S_8 = 0.762 \pm 0.012$	$S_8 = \mathbf{0.762}$

6 Explicit Falsification Criteria

To maintain rigorous scientific standards, the crisis resolutions in ‘UST-K5’ are subject to strict empirical falsification bounds.

Proposition 1 (Falsification Criteria for UST-K5). *The substrate crisis resolution model shall be considered falsified if any of the following observational conditions are established:*

1. **Convergence of Local and Early H_0 :** *If future local distance-ladder measurements (e.g., Roman Space Telescope Cepheid surveys) converge down to 67.4 km/s/Mpc, proving that the 5σ H_0 discrepancy was purely an observational systematic error.*

2. **Discovery of an Absolute Redshift Cutoff ($z_{\max}$):** If deep-field infrared surveys establish a hard physical redshift cutoff ($z_{\max} < 20$) where no luminous baryonic structures exist, contradicting a stationary, non-singular substrate.
3. **Breakdown of Pressure-Buffered S_8 Scaling:** If ultra-large-scale structure surveys (e.g., *Euclid*, *Rubin Observatory LSST*) prove that small-scale matter clustering $\sigma_8(R)$ shows zero acoustic pressure damping across $R < 10$ Mpc scales.

7 Conclusion

The Unified Substrate Theory demonstrates that the three major observational crises undermining standard Λ CDM cosmology—the 5σ H_0 tension, JWST high-redshift mature galaxies ($z > 10$), and the S_8 structure growth anomaly—are artifacts of fitting a stationary continuum medium into an expanding FLRW metric. By resolving all three crises ab-initio with zero free parameters ($k = 0$), UST-K5 establishes an internally consistent, empirically testable, and fully falsifiable framework extending from sub-atomic topological solitons (‘Paper I’) to the farthest reaches of observational cosmology.

References

- [1] R. Beem, *UST-K4: Substrate Cosmology IV: The Cosmic Microwave Background Standing Wave Spectrum, Acoustic Peaks, and Non-Linear Restoring Dynamics*, UST Cosmology Branch (2026).
- [2] R. Beem, *UST-K3: Substrate Cosmology III: Cluster Dynamics, Gravitational Lensing Refraction, and Acoustic Relaxation Lags*, UST Cosmology Branch (2026).
- [3] R. Beem, *UST-K2: Substrate Cosmology II: Galactic Dynamics, First-Principles MOND Scale Derivation, and SPARC BTFR Benchmarks*, UST Cosmology Branch (2026).
- [4] R. Beem, *UST-K1: Substrate Cosmology I: Non-Expanding Damping Redshift, Pantheon+ SNe Ia Fit, and Supernova Time Dilation*, UST Cosmology Branch (2026).
- [5] R. Beem, *Substrate Molecular Self-Organization*, UST Paper IX (2026).
- [6] R. Beem, *Hydrostatic Pressure-Shadow Gravitation and G Derivation*, UST Paper IV (2026).
- [7] I. Labbé *et al.*, *A population of red candidate massive galaxies ~ 500 – 700 Myr after the Big Bang*, *Nature* **616**, 266 (2023).